Weekly Business Review

Olist Brazilian E-commerce Dataset | Pipeline, Data Quality & Business Insights

1. Executive Summary

This review combines a repeatable Python data pipeline, SQL data-quality checks, an order-level analytical mart, and business analysis. The objective is to make order and customer performance easier to monitor while making data issues visible before results are used for decisions.

The generated metrics report a **Monthly Delivered GMV of 838576.64**, **repeat purchase rate of 3.12%**, **average review score of 4.16 / 5**, and **late delivery rate of 8.11%**. These values are a baseline; trend and segment analysis should be used to explain what is driving changes.

2. North Star and Supporting Metrics

Metric	Current result	Definition / business use
North Star: Monthly Delivered GMV	838576.64	Sum of item prices for delivered orders, grouped by purchase month; excludes freight and is not profit.
Repeat Purchase Rate	3.12%	Unique customers with more than one order divided by all unique customers; retention indicator.
Average Review Score	4.16	Average review score for delivered orders; customer-experience indicator.
Late Delivery Rate	8.11%	Delivered orders later than estimate divided by delivered orders with known delay; reliability indicator.

Interpretation: Repeat purchasing and late delivery are useful areas to monitor. These aggregate metrics alone do not establish causes; compare them across time, states, categories, and delivery groups before recommending specific interventions.

3. Revenue and Order Trends

Monthly delivered GMV tracks the value of products in successfully delivered orders. Read it alongside delivered-order volume: GMV can rise because more orders are delivered, because the average basket value increases, or both.

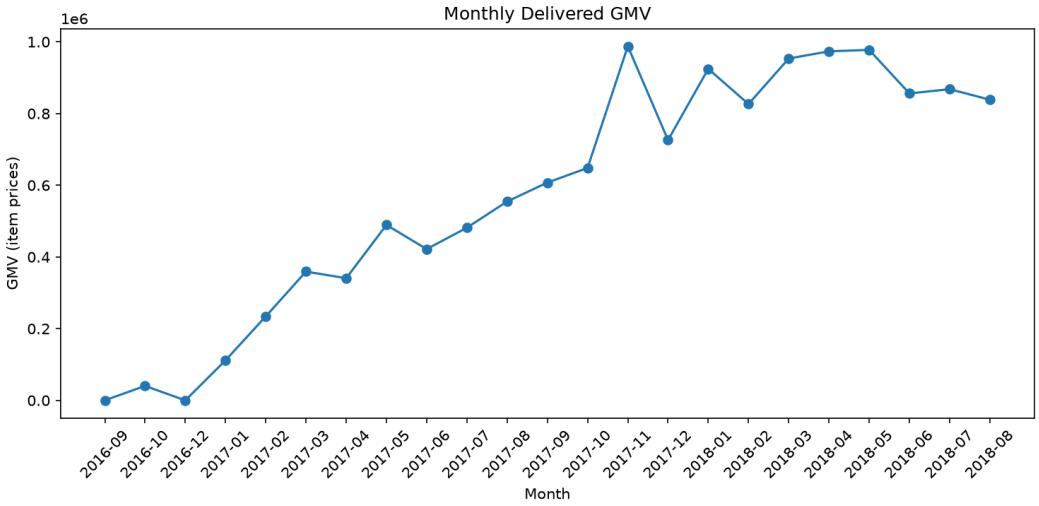


Figure 1. Monthly delivered GMV. Look for month-over-month increases, declines, and recurring peaks; validate unusual changes against order volume.

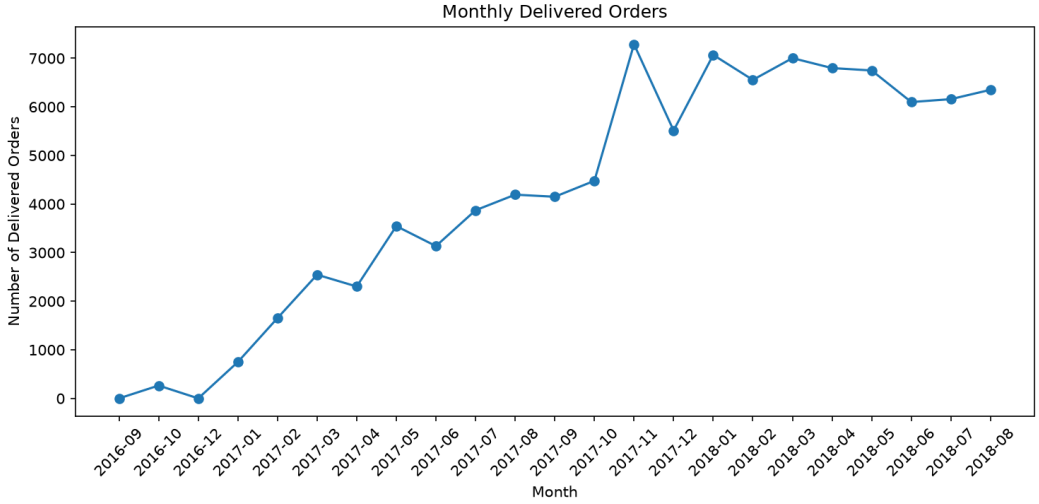


Figure 2. Delivered orders by month. Comparing this series with GMV helps distinguish order-volume changes from changes in value per order.

Questions for the weekly review

- Which months show the largest GMV change?
- Do order counts move in the same direction as GMV?
- Are peaks concentrated in particular periods or categories?

4. Customer Experience and Delivery

The average review score summarises feedback, while the late-delivery rate measures the share of eligible delivered orders that arrived after their estimated date. Review scores can be compared across early, on-time, and late delivery groups to identify an association, not to prove causation.

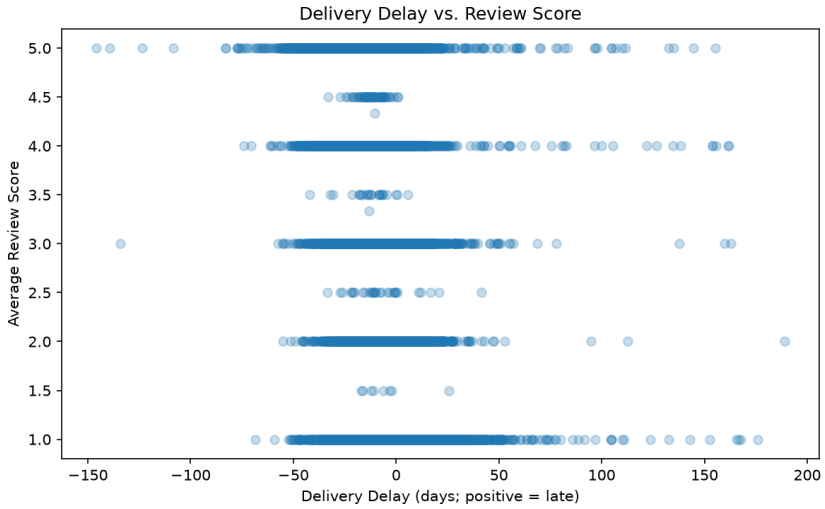


Figure 3. Delivery timing and review score. Check whether average review scores differ by delivery category and whether group sizes are comparable.

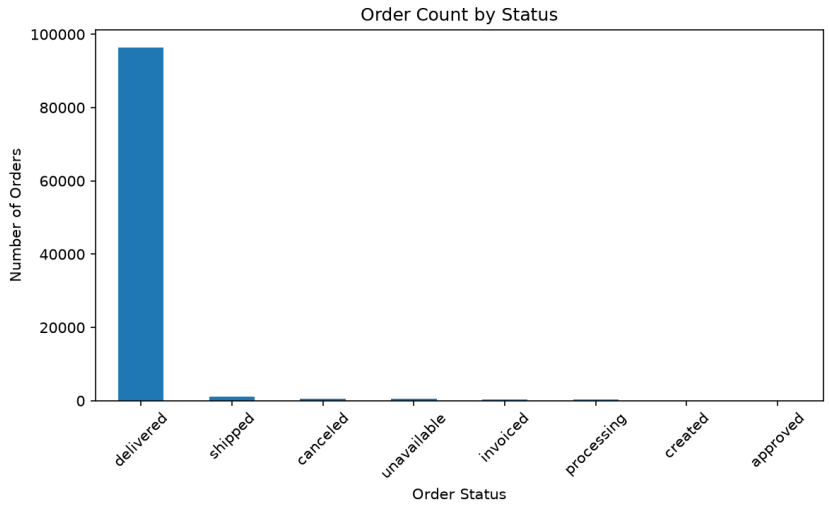


Figure 4. Order-status distribution. This provides context on how many records reached delivery versus other order states.

Practical actions

- Investigate late deliveries by state, seller, and product category.
- Compare review-score distributions—not only averages—across delivery groups.
- Monitor status mix and missing delivery timestamps to understand metric coverage.

5. Data Pipeline and Data Quality

Pipeline stages: Raw CSV ingestion → cleaned staging tables → SQL quality checks → order-level mart → business metrics and charts. The mart is designed to hold one row per order, combining item revenue, freight, payment totals, review score, and delivery timing.

Idempotency and reliability: Raw ingestion performs a refresh from source CSV files; staging and mart tables are rebuilt from those inputs. The pipeline logs progress and raises an error when a configured quality check fails. Re-running should produce the same result when the source files and logic have not changed.

Quality checks: missing order IDs, duplicate order IDs, invalid review scores, negative item prices, orders with missing customers, and delivery dates before purchase. Review the generated quality-check output after each run; report the actual failure counts and handling decisions from the final execution.

6. SQL Analysis

Monthly revenue and growth: aggregates delivered-order revenue by month and uses LAG to compare against the previous month.
Top categories by state: aggregates category revenue by customer state and uses DENSE_RANK to return the top three categories per state.
Rolling daily orders: uses a window over the current date and six preceding rows; confirm missing calendar dates are filled with zero if interpreting it as a seven-day window.
Repeat purchases: uses customer_unique_id to count distinct customers with repeat orders.
Delivery and reviews: compares delivery delay categories with average review scores.

7. Recommendations

1. Improve delivery reliability. Segment late deliveries by state, seller, and product category; prioritise the segments with the largest volume and highest delay rate.

2. Improve customer retention. Track repeat purchase rate over time and test post-purchase engagement or targeted re-engagement campaigns. Compare results against a baseline.

3. Protect customer experience. Review ratings for late versus on-time deliveries and investigate whether specific locations or categories account for weaker scores.

4. Interpret growth with context. Track GMV with delivered orders and, where available, average order value. Avoid attributing growth to one factor without checking the related measures.

8. Limitations and Reporting Notes

This is a descriptive analysis of historical marketplace records. Associations between delivery timing and reviews do not prove causation. GMV is based on item prices and should not be described as profit; confirm the reporting period and currency label before presenting the value externally. Metrics depend on their stated denominators and on the completeness of timestamps and source records.

Conclusion

The pipeline and SQL analyses provide a reusable baseline for marketplace monitoring. The next review should compare metrics with the previous period, investigate the strongest segment-level changes, and document any data-quality exceptions before business decisions are made.

Source: Olist Brazilian E-commerce dataset and outputs generated by this project.